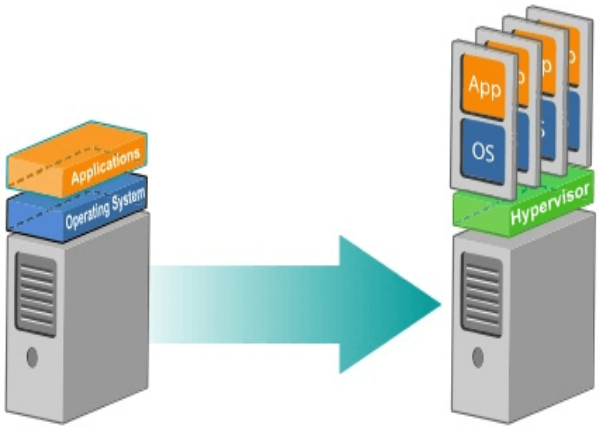


CLOUD COMPUTING COURSE

Lecture9-CPU Virtualization Types

ABSTRACT
Lecture9 Introduces CPU virtualization types.

Instructor: Asma'a Khtoom

CPU Virtualization

OS Modes

There are two modes in which a modern OS works:

1. User mode

2. Kernel mode.

Kernel is the core of the OS. It is responsible for allocating of various types of computer resources like memory, CPU, devices etc. and many other tasks related to functioning of OS

Why? To ensure protection and security of the System from unauthorized or errant users.

User Mode Vs Kernel Mode

- Whenever the system boots, OS transfers control to kernel mode, where kernel runs all the necessary routines and then kernel switches to user mode.

- User mode allows users to run any programs of their choice.

CPU Instruction Types:

Operating system have two types of instructions:

1. **User instructions:** instructions, which normal users can perform. User instructions are those that are not privileged.
   Examples:

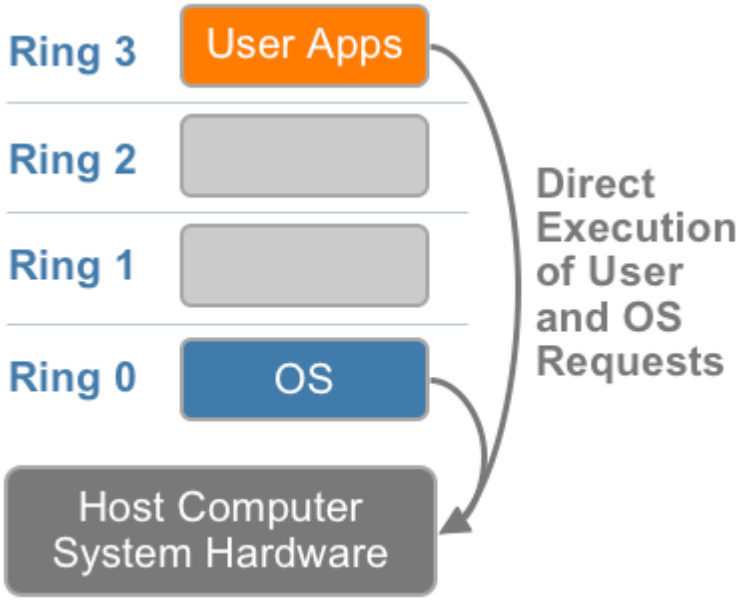
   1. Reading the status of Processor
   2. Reading the System Time
   3. Sending the final printout of Printer

2. **Privileged instructions:** These instructions are allowed to be run only in **kernel mode** of the OS. They are any instructions that could alter some parts of OS or any of your resources, IO systems. Kernel executes such instructions and returns control to User. **Examples:**

   1. I/O instructions.
   2. Clear the Memory or Remove a process from the Memory

Protection Rings

The simple security models in a computer call for rings of protection.



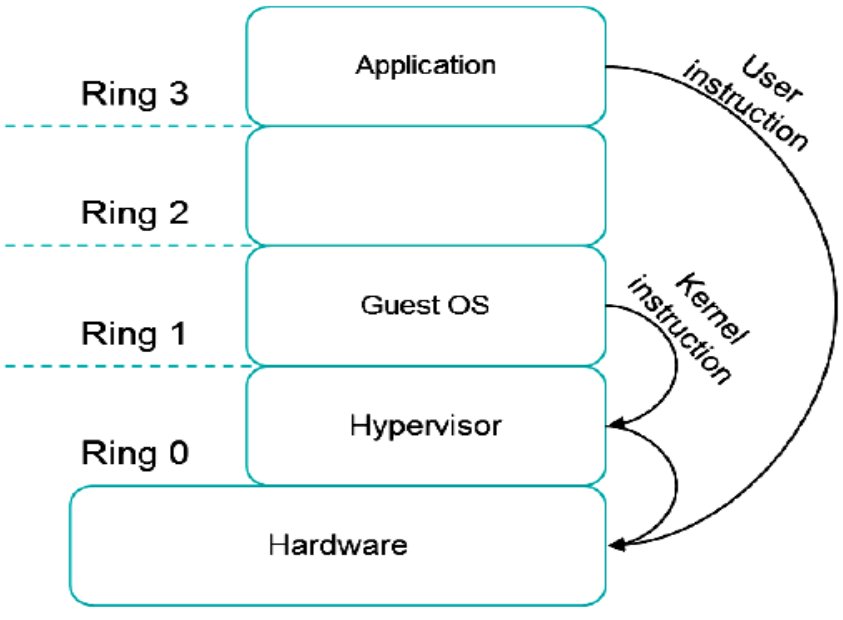
CPU Virtualization Types

CPU virtualization can be achieved by following techniques:

- **Full virtualization** using binary translation
- **Para virtualization**
- **Hardware assisted virtualization**

Full virtualization:

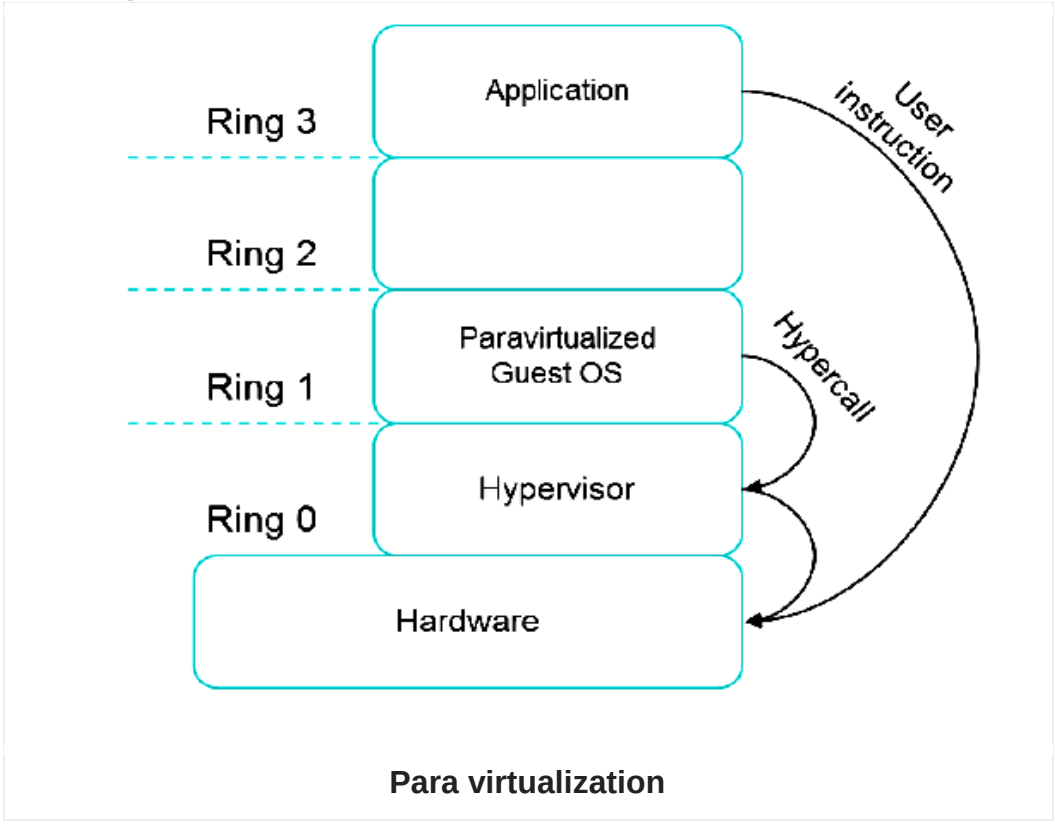
- Hypervisor directly interact with the hardware such as CPU,disks.
- Does dynamic binary translation/ rewriting
- Creates emulation layer to emulate hardware
- Virtual Machine Monitor (Hypervisor) traps and emulates privileged instructions.
- The guest OS is not aware it is being virtualized and requires no modification.
- It offers the best isolation and security for virtual machines.
- It simplifies migration and portability.
- Full virtualization is usually bit slower, because of all emulation.
- Examples- Hyper-V, VMware vSphere, QEMU.



Full virtualization

Para virtualization:

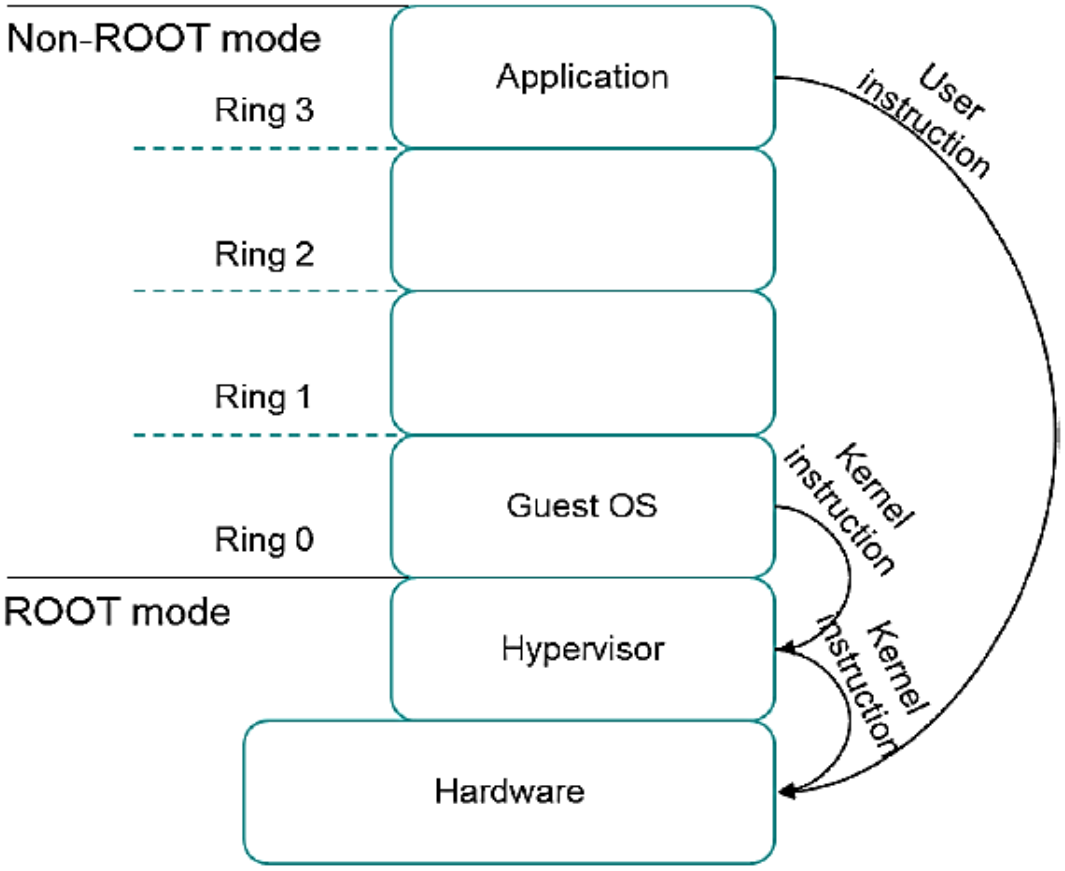
- **The guest operating system is aware that it is a guest.**
- **Guest OS is modified to process privileged instructions.**
- **Hypervisor does not need large amounts of processing power to manage guest OS.**
- **VMM is no longer required to translate instructions; it is only responsible for handling virtualization layer.**
- **Guest OS uses specialized API to talk with VMM.**
- **The performance of Para virtualization is more than full virtualization**
- **Para virtualization cannot support unmodified operating systems (e.g. Windows 2000/XP), its compatibility and portability is poor.**
- **Example- Xen**



Para virtualization

Hardware-Assisted virtualization:

- **The VMM uses processor extension (Intel-VTx or AMD-V) to intercept and emulate privileged instructions.**
- **Guest OS runs at kernel level (Ring 0) and VMM runs in more privileged Ring than the Ring 0.**
- **Hardware-assisted virtualization is used to solve virtualization vulnerabilities, simplify VMM software, and eliminate the need for paravirtualization or binary translation.**



Hardware Assisted Virtualization